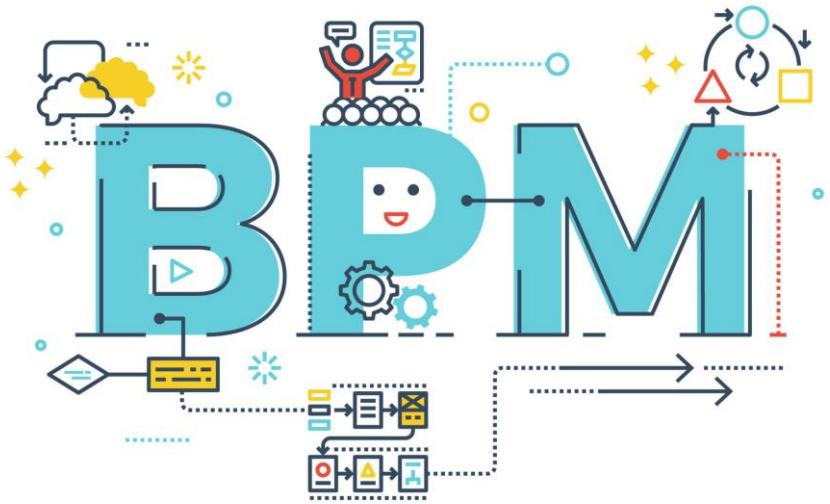
Business Process Management

Lab 1: Process Identification



What is a Business Process (BP) ?

DEFINITION :

- **A business process :** set of linked activities performed by people or systems produces an outcome that contributes **to business goals**.
- **Processes** are everywhere in a business.

→ From onboarding new employees to exit interviews, vendor invoicing, and sales, every business outcome is a result of repeatable tasks and activities.

What is a Business Process (BP) ?

EXAMPLE (1) :

Process of procurement

To purchase the materials you need to follow these steps:

- identifying a product’s need.
- Contact the purchasing department
- The purchasing department reviews procurement requests.
- A budget approval request with all financial details is sent to the financial department.
- When a request is approved, the purchasing department sends it out to suppliers
- The vendor is then selected and a contract is signed.
- Upon receiving the goods, the purchasers inspect the quality of delivery.
- Orders and invoice slips must match for invoice payment to be approved.

What is a Business Process (BP) ?

EXAMPLE (2) :

Sales process

The process of making a sale is fundamental across all industries, In most businesses , the following steps are followed:

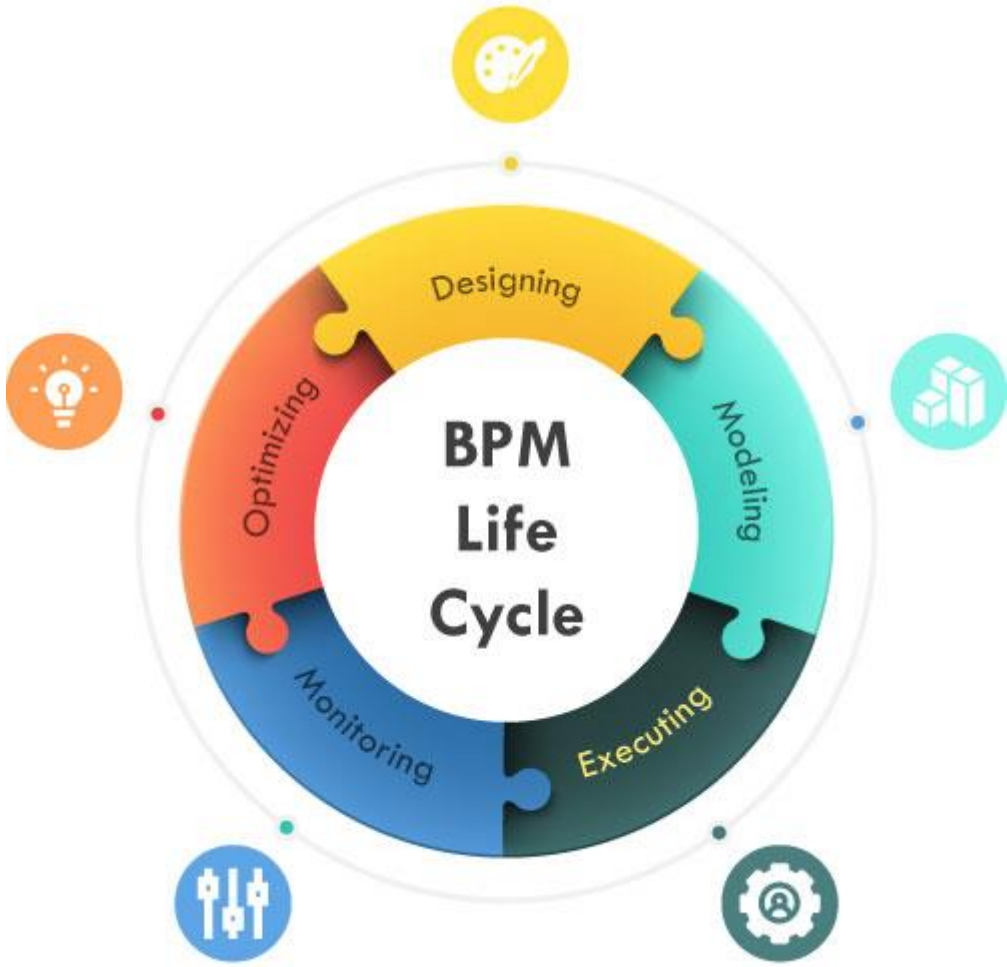
- Proposal sharing
- Quotation sending
- The negotiation process
- Ordering products/services
- Maintaining sales records
- Product or service delivery
- Post-Sale Follow-up
- The billing process
- The payment process
- Renewals and Retention

What is a Business Process Management (BPM) ?

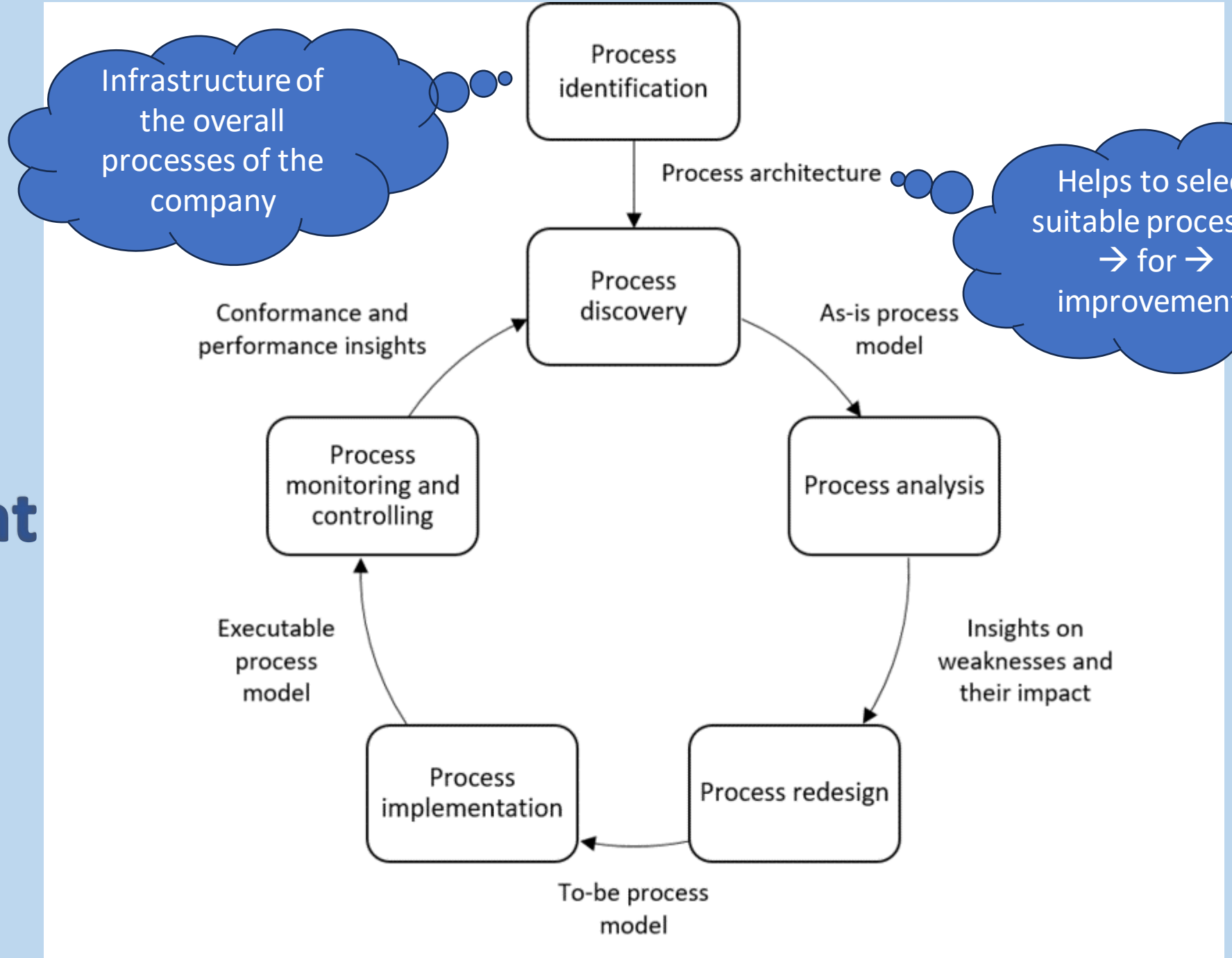
DEFINITION :

- **Business <u>process management</u> (BPM) :**

 is a systematic <u>approach</u> to improving those processes, which helps organizations achieve their **business goals**.



Business Process Management Lifecycle



Business Strategy

- **Process Identification** ← Related to → **Business Strategy**
- **Business strategy** is an organizational perspective on setting and meeting business goals. (Mintzberg)
- It clears the direction of BPM initiatives in a company
- **Process Identification** → helps to operationalize this **Business Strategy**

Process Identification

- It refers to those management activities that aim to :

  - Define the set of business processes of an organization systematically .

and

  - Establish clear criteria for selecting specific processes for improvement.

- The reason for conducting process identification is that an organization should focus on those processes that either :

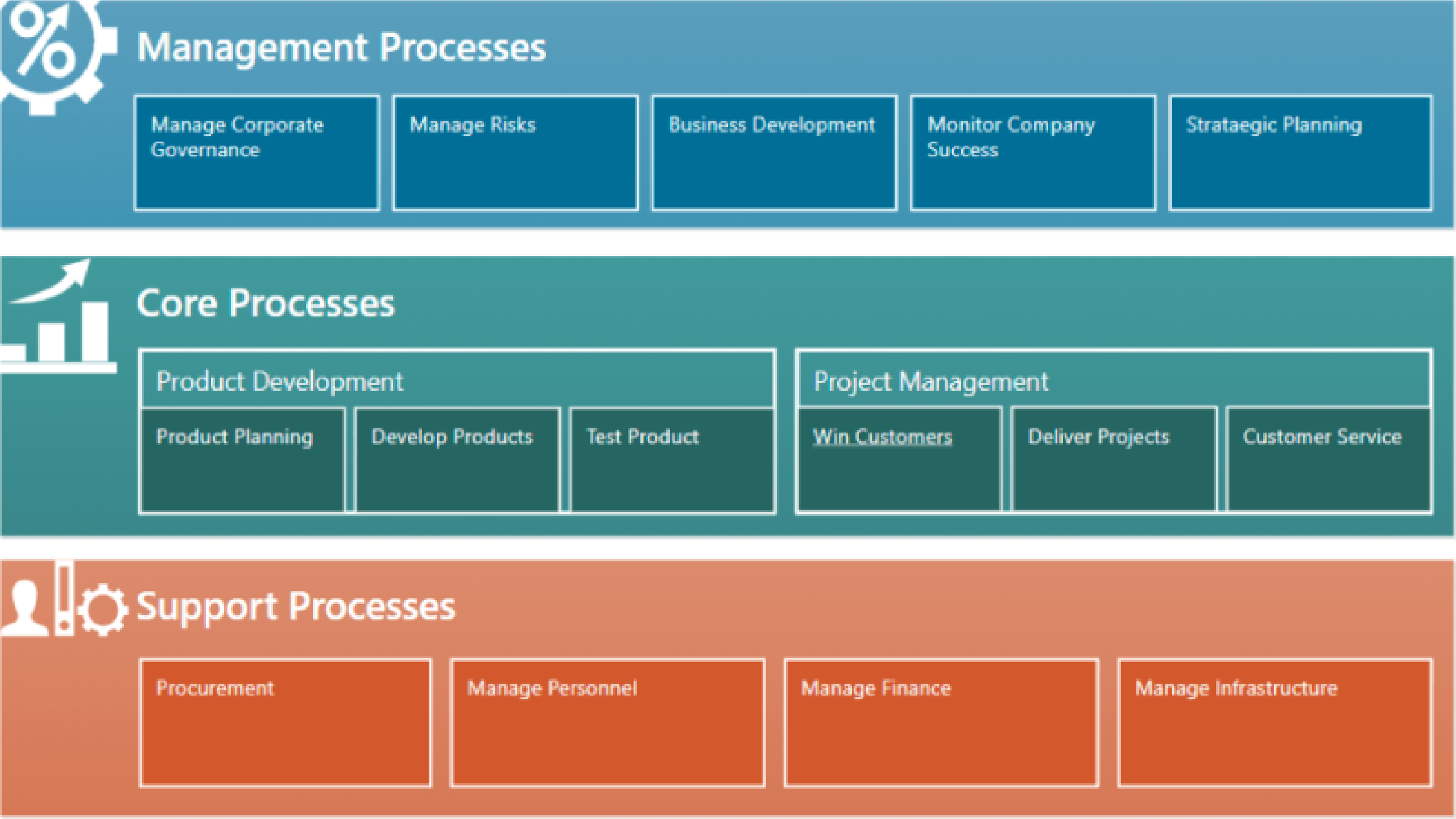
  - create value of strategic relevance

or

  - that have substantial problems (or both).

Process Architecture

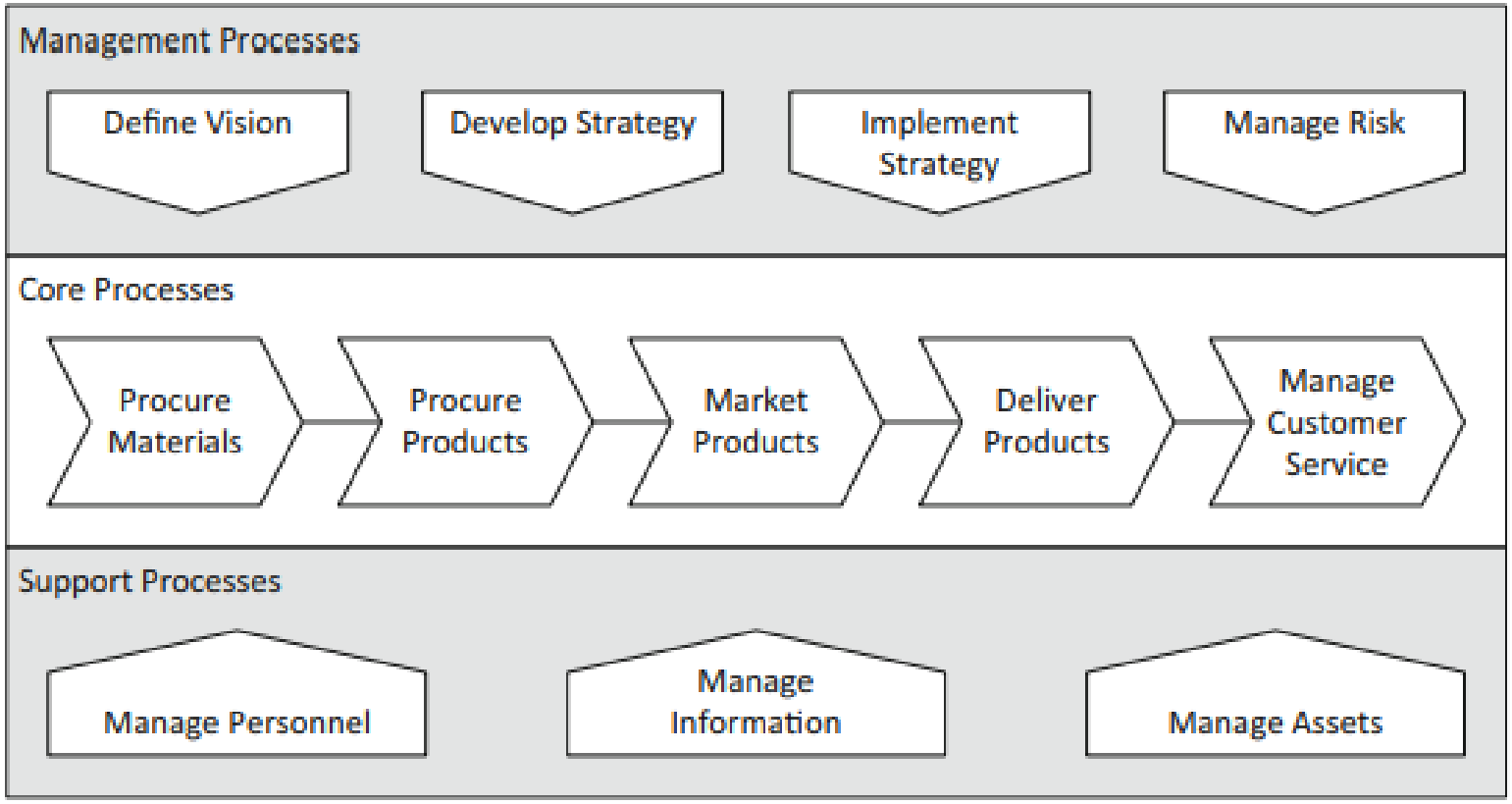
- **Process Landscape Model** which describes the most **abstract** view of the process architecture.
- **The main purpose** of a process landscape is to specify corporate processes that are directly involved in adding value to the company.
- If such processes are linked to each other as a sequence of functions, they form a **process landscape**.
- The **process landscape model** requires the involvement of major stakeholders of the organization, either using *interviews* or, preferably, using a *workshop*


Management Processes
Manage Corporate Governance
Manage Risks
Business Development
Monitor Company Success
Strataegic Planning
Core Processes
Product Development
Product Planning
Develop Products
Test Product
Project Management
Win Customers
Deliver Projects
Customer Service
Support Processes
Procurement
Manage Personnel
Manage Finance
Manage Infrastructure

Process Landscape Model

- **Process Landscape Model** which describes the most abstract view of the process architecture.

- Example**: Process Landscape Model for a Production Company**



Process Categories & Examples

Management processes determine how the company is run. In other words, they are the directions, rules, and practices for the core and support processes.

- A typical example is defining corporate strategy.

Core processes directly create value for a customer.

- For example, in a manufacturing company, the production process is a core process.

Support processes enable the execution of the core processes.

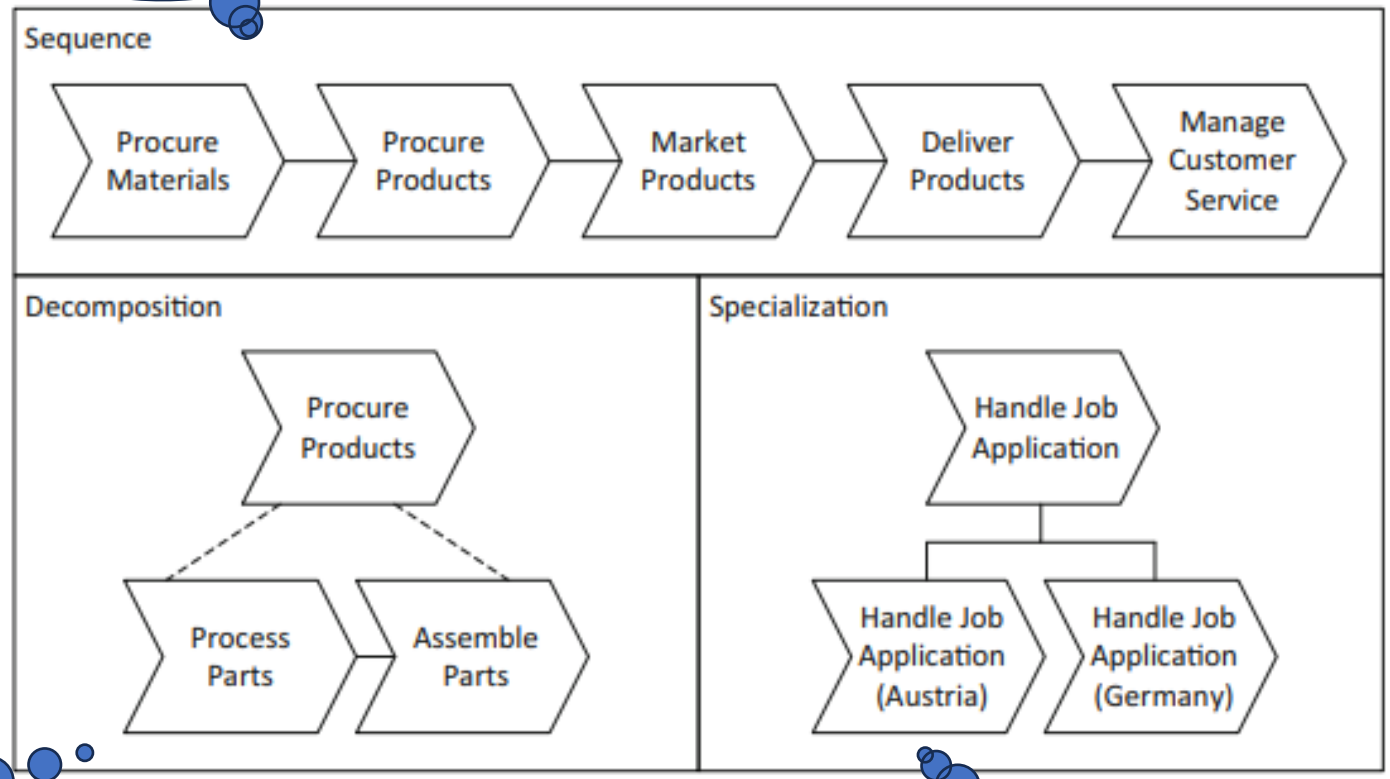
- Typical examples are human resource management, IT management, and invoicing.

Business process Categories & Examples

- **Examples of Management processes include the following:**
  - internal communications
  - governance
  - strategic planning
  - budgeting
  - infrastructure or capacity management
- **Examples of Core Processes include the following:**
  - taking customer orders
  - processing product payments
  - managing bank accounts
- **Examples of Supporting processes include the following:**
  - accounting
  - HR management
  - workplace safety

Relationships Between Processes

Express that activities are in the sequential order



Identify a major overarching activity & we decompose it into its parts

Activity that have different ways of doing it

Example (1)

Develop a Process Landscape for the following case study:
An error management system in a vet clinical laboratory

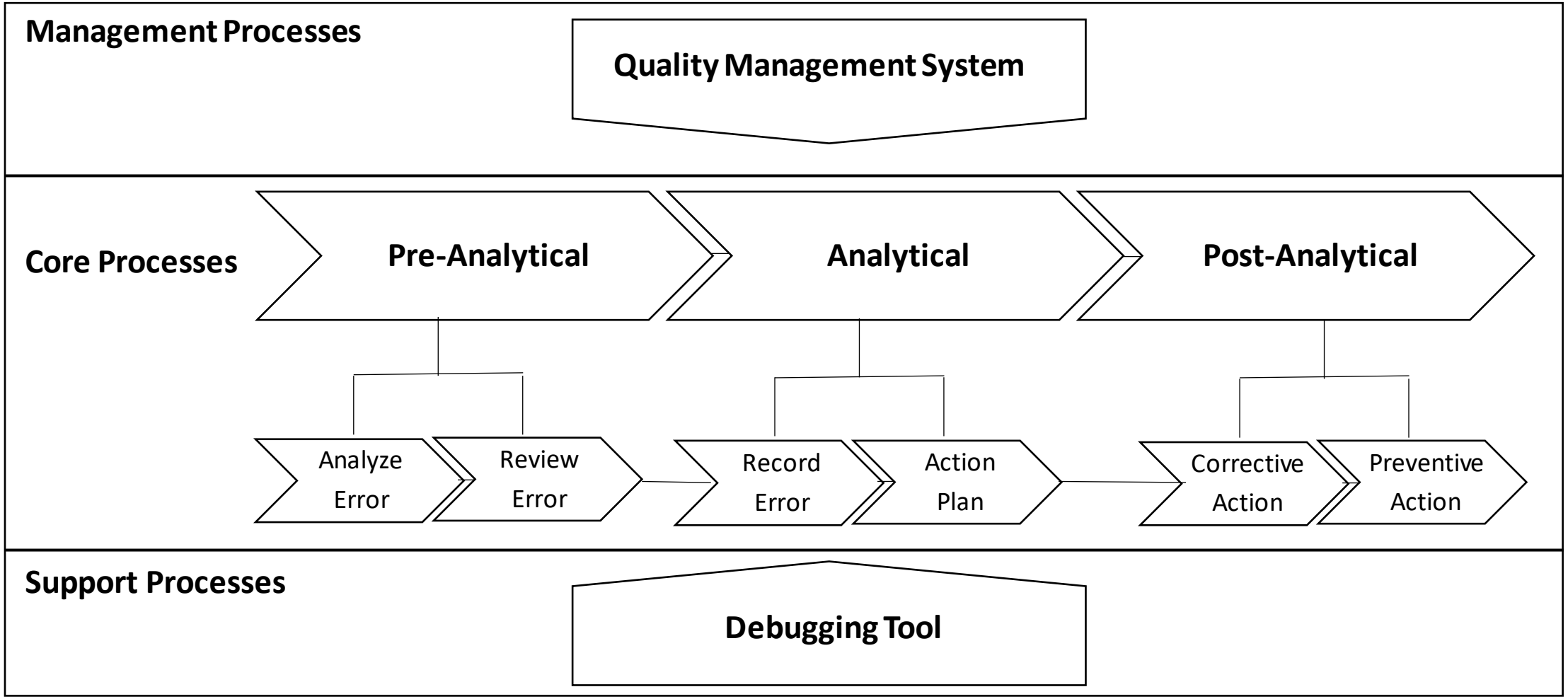
- Error recording and management is an integral part of a clinical laboratory quality management system. Analysis and review of recorded errors lead to corrective and preventive actions through modification of existing processes and, ultimately, to quality improvement. Laboratory errors can be divided into pre-analytical, analytical, and post-analytical errors depending on where in the laboratory cycle the errors occur.
- The purpose of the current report is to introduce an error management system in use in a vet diagnostic laboratory as well as to examine the amount and types of errors recorded during the 8-year period from 2003 to 2010.

Develop a Process Landscape for the following case study:
An error management system in a vet clinical laboratory

- Annual error reports generated during this period by the error recording system were reviewed, and annual error rates were calculated. In addition, errors were divided into pre-analytical, analytical, post-analytical, and “other” categories, and their frequency was examined. Data were further compared to that available from human diagnostic laboratories.
- Finally, sigma metrics were calculated for the various error categories. Annual error rates per total number of samples ranged from 1.3% in 2003 to 0.7% in 2010.
- The incremental annual reduction of error shows that the use of an error management system led to quality improvement.

Process Landscape

EXAMPLE (1) :

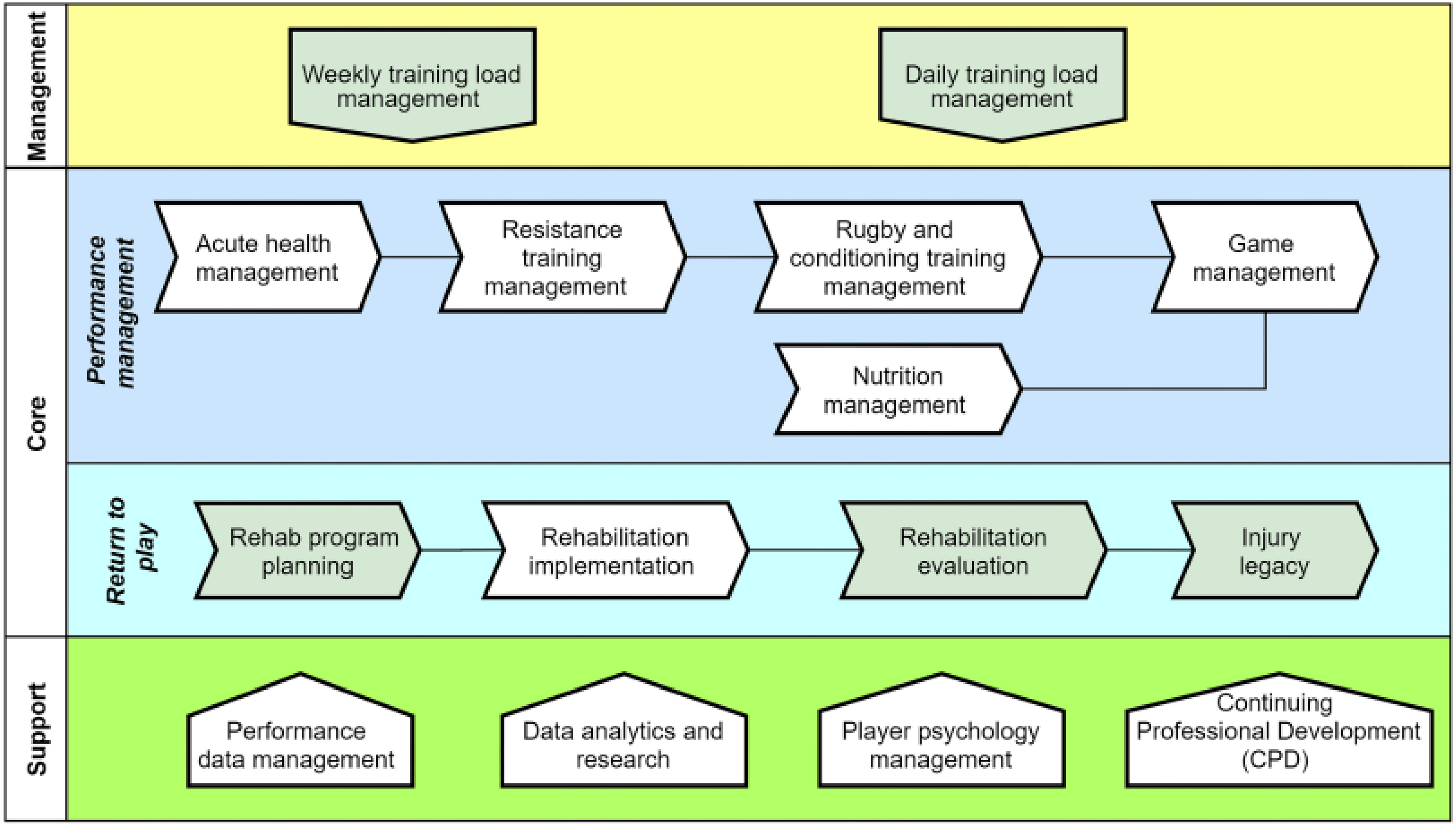


Process Profile

Name of Process: Analytical	
Vision: The objective of the analytical process is to analyze the reasons of errors and creates an action plan to solve these errors.	
Process Owner: Head of Quality Management	
Customer of Process: • Client	**Expectation of Customer:** • Complete Action Plan
Outcome: Action Plan	
Trigger: Error Detected	
First Activity: Error Recording …….. **Last Activity:** Action Plan	
Interfaces Inbound: Pre-Analytical **Interfaces Outbound:** Post-Analytical	
Required Resources: Testing Team	
Process Performance Measures: Error Detection Rate	

Example (2)

- As the first step of an optimization project in a sporting environment, this study aimed to evaluate the current state and the improvement opportunities in the player management information flow executed within the High-Performance Unit (HPU) at a professional rugby union club in England.
- Players always get injured and require a rehabilitation plan to return to play as soon as possible.
  - **Core**: Key player management execution processes in the HPU.
  - **Management**: Processes providing guidance for the execution of the core processes.
  - **Support**: Processes assisting for a smooth execution of the core and management processes


Management
Weekly training load management
Daily training load management
Core
Performance management
Acute health management
Resistance training management
Rugby and conditioning training management
Game management
Nutrition management
Return to play
Rehab program planning
Rehabilitation implementation
Rehabilitation evaluation
Injury legacy
Support
Performance data management
Data analytics and research
Player psychology management
Continuing Professional Development (CPD)

Assignment

- **<u>Page 30 Exercise 1.6 chapter 1</u>**
- Consider the following process at a pharmacy.

Customers drop off their prescriptions either in the drive-through counter or in the front counter of the pharmacy. Customers can request that their prescription be filled immediately. In this case, they have to wait between 15 min and 1 h depending on the current workload. However, most customers are not willing to wait that long, so they opt to nominate a pickup time at a later point during the day. Generally, customers drop their prescriptions in the morning before going to work (or at lunchtime) and they come back to pick up the drugs after work, typically between 5 p.m. and 6 p.m. When a prescription is dropped off, a technician asks the customer for the pick-up time and puts the prescription in a box labeled with the hour preceding the pick-up time. For example, if the customer asks to have the prescription be ready at 5 p.m., the technician will drop it in the box with the label 4 p.m. (there is one box for each hour of the day). Every hour, one of the pharmacy technicians picks up the prescriptions due to be filled in the current hour. The technician then enters the details of each prescription (e.g., doctor details, patient details and medication details) into the pharmacy system. As soon as the details of a prescription are entered, the pharmacy system performs an automated check called Drug Utilization Review (DUR).

This check is meant to determine if the prescription contains any drugs that may be incompatible with other drugs that had been dispensed to the same customer in the past, or drugs that may be inappropriate for the customer taking into account the customer data maintained in the system (e.g., age). Any alarms raised during the automated DUR are reviewed by a pharmacist who performs a more thorough check. In some cases, the pharmacist even has to call the doctor who issued the prescription in order to confirm it.

After the DUR, the system performs an insurance check in order to determine whether the customer's insurance policy will pay for part or for the whole cost of the drugs. In most cases, the output of this check is that the insurance company will only pay for a certain percentage of the costs, while the customer has to pay for the remaining part (also called the co-payment). The rules for determining how much the insurance company will pay and how much the customer has to pay are very complicated. Every insurance company has different rules. In some cases, the insurance policy does not cover one or several drugs in a prescription, but the drug in question can be replaced by another drug that is covered by the insurance policy.

When such cases are detected, the pharmacist generally calls the doctor and potentially also the patient to determine if it is possible to perform the drug replacement. Once the prescription passes the insurance check, it is assigned to a technician who collects the drugs from the shelves and puts them in a bag with the prescription stapled to it. After the technician has filled a given prescription, the bag is passed to the pharmacist who double-checks that the prescription has been filled correctly. After this quality check, the pharmacist seals the bag and puts it in the pick-up area. When a customer arrives to pick up a prescription, a technician retrieves the prescription and asks the customer for payment in case the drugs in the prescription are not fully covered by the customer's insurance.

- **With respect to the above process, consider the following questions:**
  1. **What type of process is the above one: order-to-cash, procure-to-pay, application-to-approval, or issue-to-resolution?**
  2. **Who are the actors in this process?**
  3. **Who are the customers?**

- **Assignment individually:**

1. **Create Process Landscape**
2. **For one of core processes, create a process profile**

Thank you